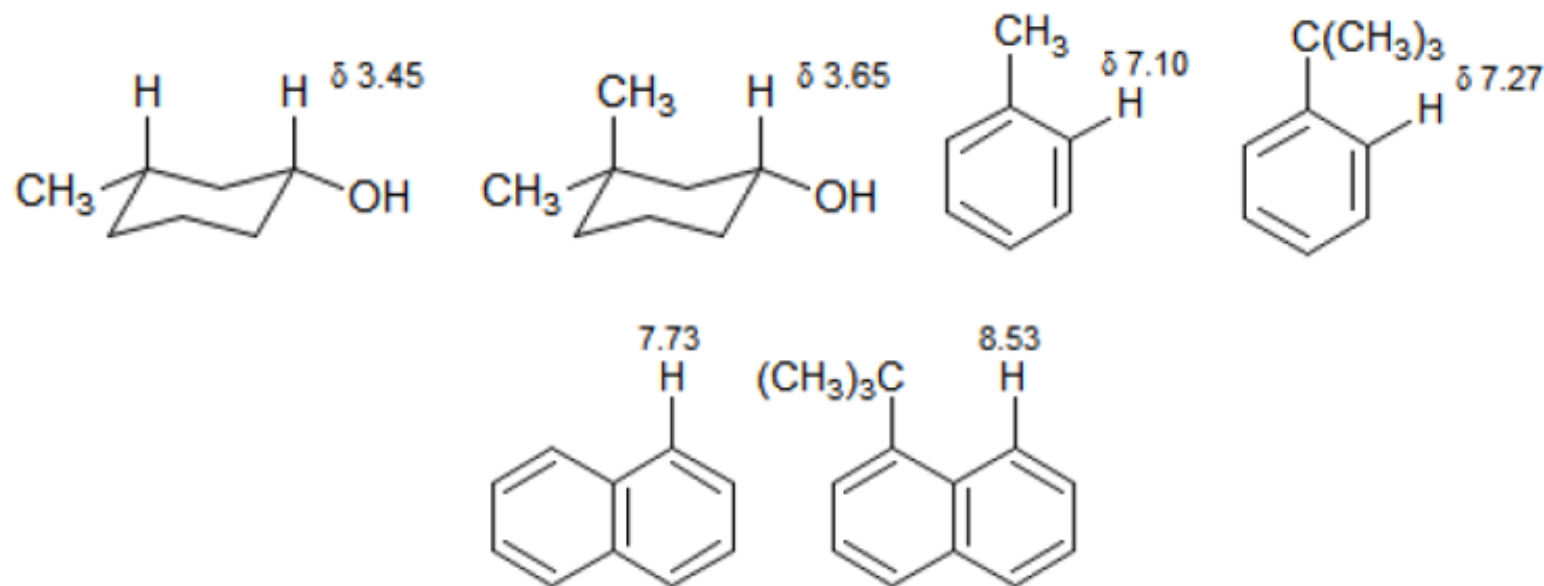


Steric effects:

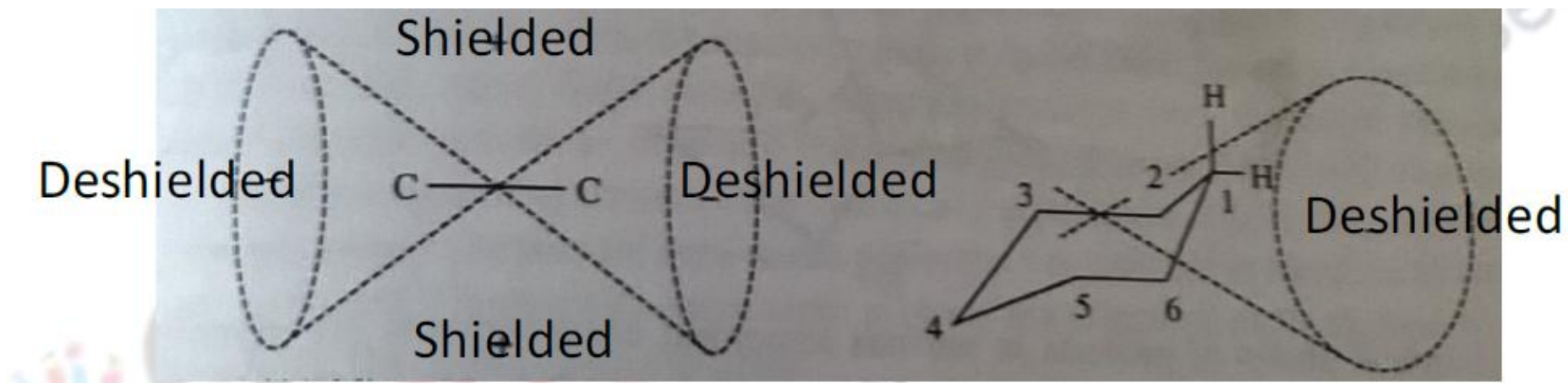
In overcrowded molecules, it is possible that some protons may occupy sterically hindered position resulting in van der Waal's repulsion. In such molecules the electron cloud of a bulky group (hindering group) will tend to repel the electron cloud surrounding the proton. Thus, such a proton will be deshielded and will resonate at slightly higher value of chemical shift than the expected in the absence of this effect.



Alkanes

The sigma bonding electrons also produce anisotropic effects but it is less pronounced than that produced by circulating π -electrons. The axis of the C-C bond is the axis of the deshielding. The single bond anisotropy can be observed occurs in the alkanes of type RCH_3 , R_2CH_2 , R_3CH_3 as well as the equatorial versus axial protons in a fixed cyclohexane ring.

The equatorial protons in a conformationally rigid cyclohexane ring appear more downfield by nearly 0.5 ppm than the axial protons on the same carbon. This effect can be explained on the basis of anisotropy effect shown in the figure, where the equatorial protons reside in the deshielding ($+\delta$) region of the C-C anisotropy, and the axial in the shielding ($-\delta$) region. However, the C1 axial proton is within the shielding region of the C2-C3 and C5-C6 bonds, whereas C1 equatorial proton is within their deshielding region. The **methyl** protons (CH_3) appear near 0.9 ppm, a **methylene** protons (CH_2) shows signal near 1.2 ppm, and a **methine** proton (CH) shows near 1.6 ppm. The high frequency position of methine protons (R_3CH) compared with methylene protons (R_2CH_2) and that of compared with methyl protons $\text{R}-\text{CH}_3$) may be attributed to the anisotropy of the additional C-C bonds.



Mass Spectrometry

Mass spectrometry

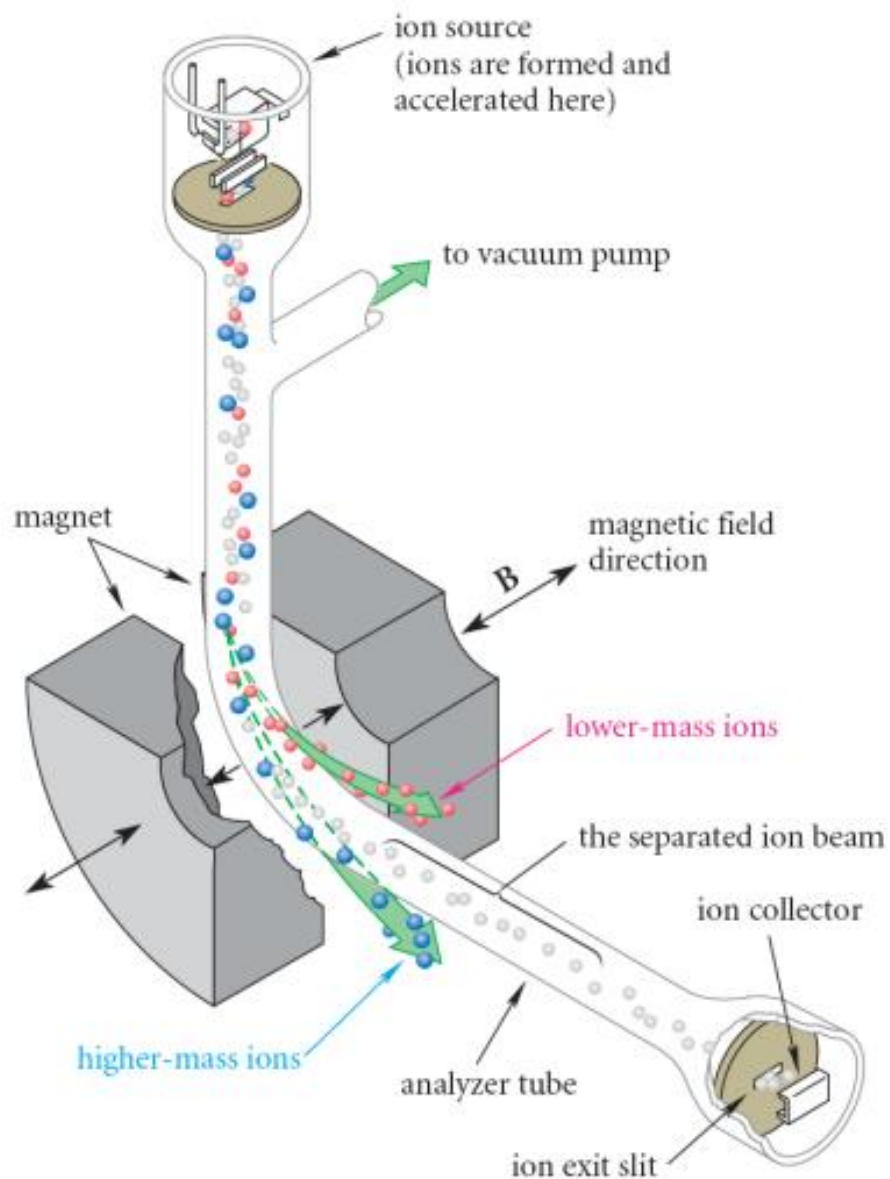
Mass spectrometry weighs the molecule

It is not easy to weigh a neutral molecule, and a mass spectrometer works by measuring the mass of a charged ion instead: the charge makes the molecule controllable by an electric field. A mass spectrometer therefore has three basic components:

- something to volatilize and ionize the molecule into a beam of charged particles
- something to focus the beam so that particles of the same mass:charge ratio are separated from all others
- something to detect the particles

All spectrometers in common use operate in a high vacuum and use one of several methods to convert neutral molecules into cations, the most common being electron impact, chemical ionization, and electrospray.

Electron Impact Mass Spectrometry (EI-MS)



Electron-Impact Mass Spectrometry (EI-MS) is used to determine molecular masses and connectivity by bombarding a molecule with high energy electrons and detecting the masses of the resulting fragments.

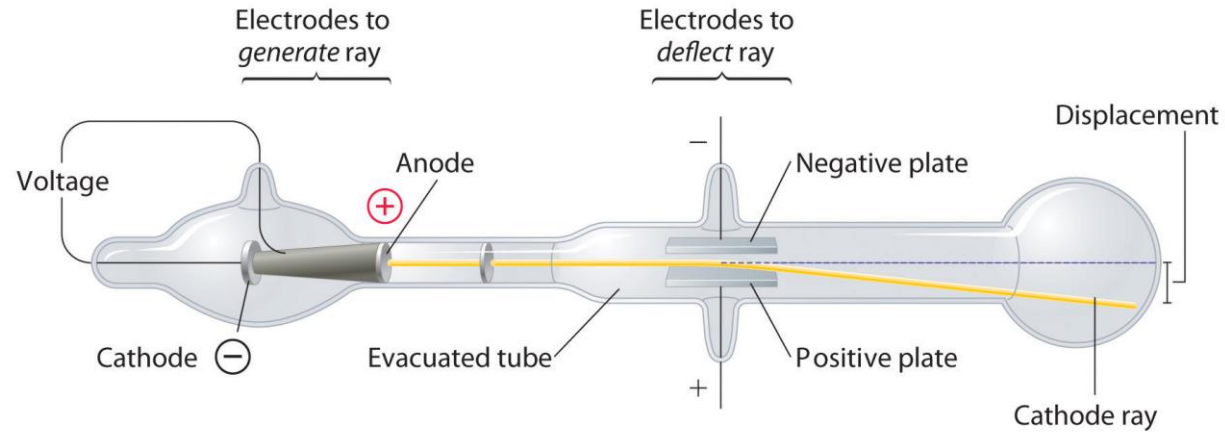
- Highly sensitive and destructive.
- Ionization from a 70 eV beam of electrons.
- Ions formed are controlled by a magnetic field, mass selected, and detected at an ion collector.
- Detection does not use electromagnetic radiation (not spectroscopy).
- Fragmentation provides structural and molecular mass information.

Ionization Methods

1. Electron ionization (EI)
2. Chemical ionization (CI)
3. Electrospray ionization (ESI)
4. Desorption Ionization

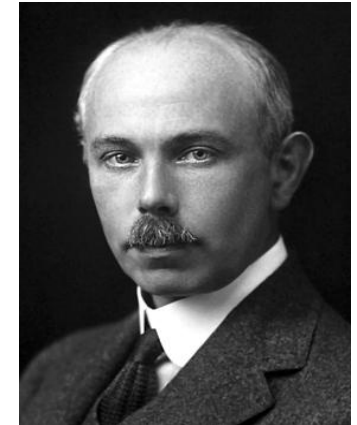
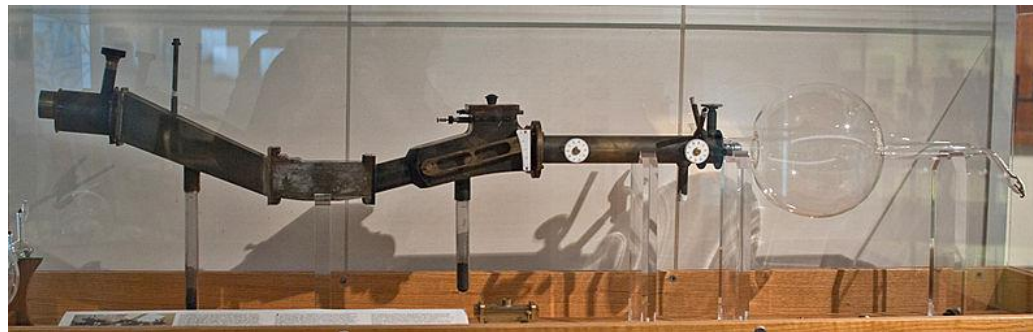
Historical Perspective

- In 1890s, J. J. Thomson determined charge-to-mass ratio of electrons



J. J. Thomson
Nobel Prize
1906

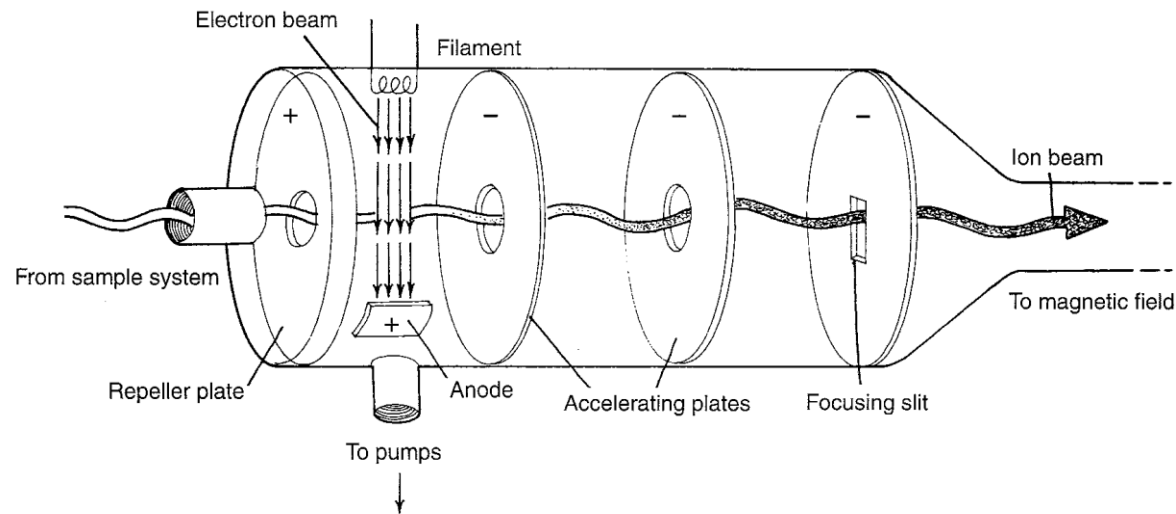
- Mass spectrometry of isotopes: Francis Aston



Francis Aston
Nobel Prize
1922

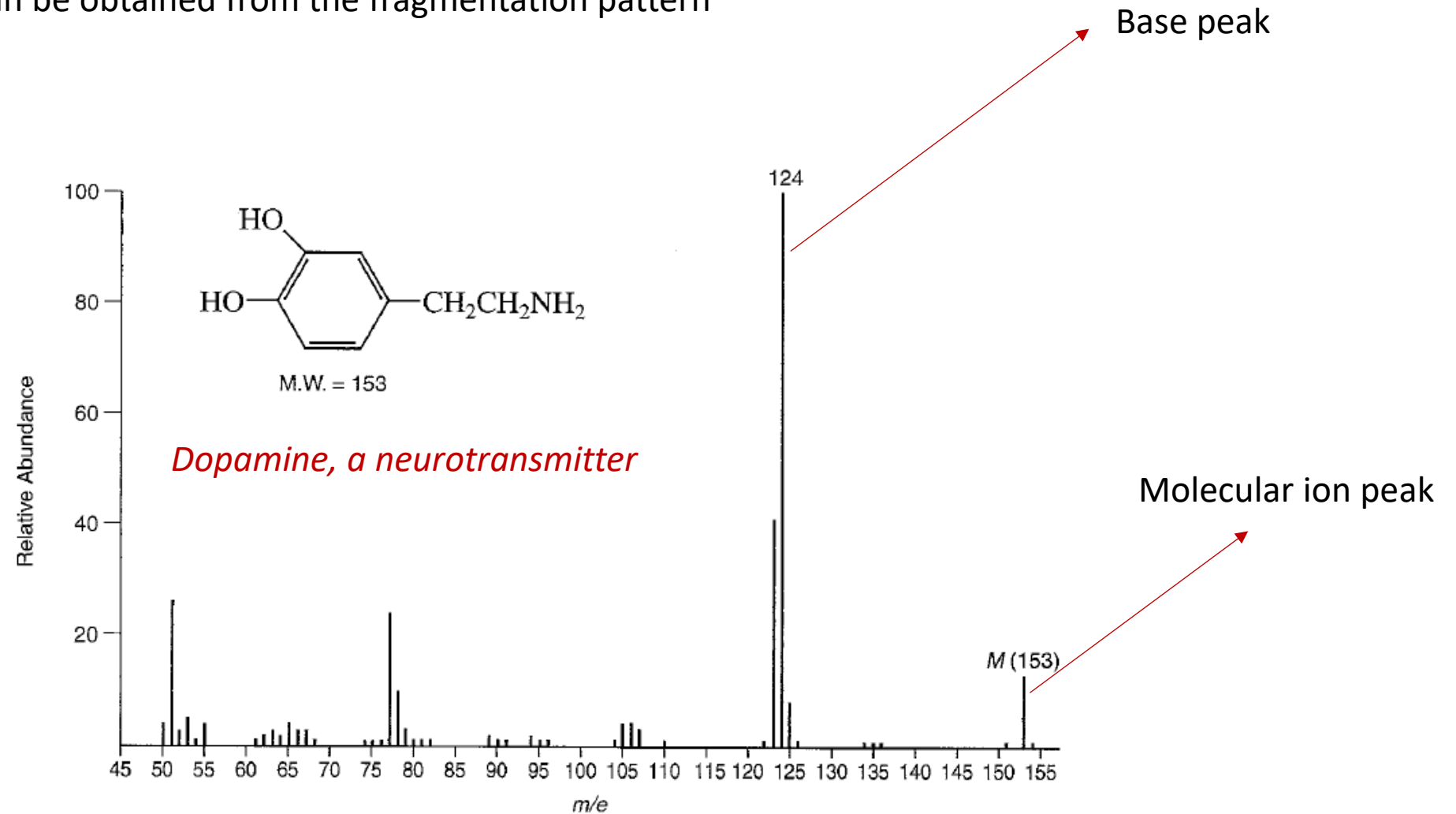
Electron Ionization (EI)

- The molecules are bombarded with high energy electrons; ionization of molecules
- Typically positive ions are produced and directed to analyser chamber by a repeller plate
- The ionization potential of organic molecules is in the range of 8 to 15 eV
- However, typically 50-70 eV electrons are used
- *Excess kinetic energy leads to the fragmentation of molecular ions*



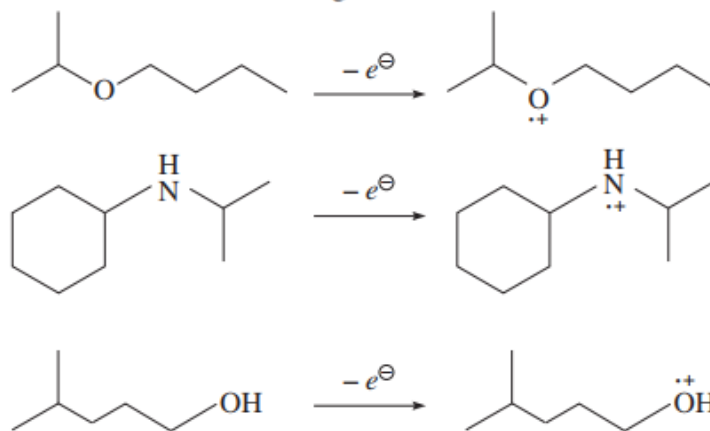
In EI-MS, radical cations are generated and analysed

- Molecular ions
- Fragmented ions
- Structural information can be obtained from the fragmentation pattern

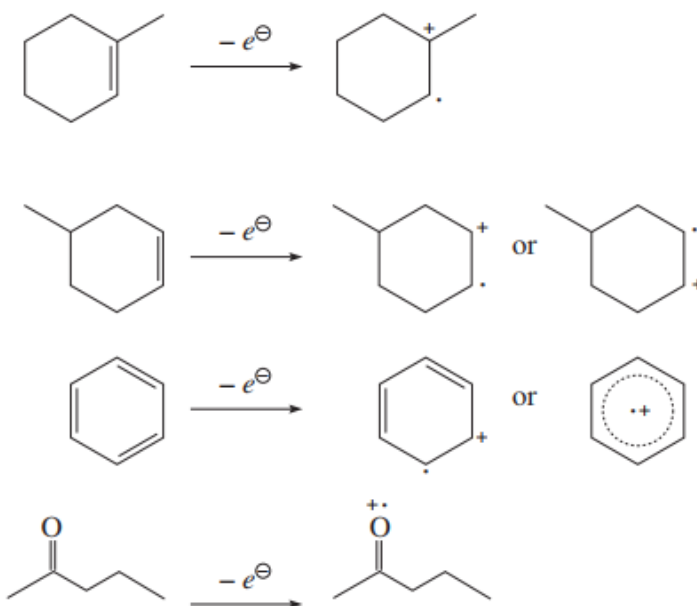


The electrons most likely to be ejected during the ionization event are the ones that are in the highest potential energy molecular orbitals, that is the electrons held least tightly by the molecule. Thus, it is easier to remove an electron from a nonbonding orbital, n , than it is to strip an electron from a p orbital. Similarly it is much easier to eject an electron from a p orbital in comparison to a s orbital. The molecular ion can be represented with either a localized or a non-localized charge site. Some examples of loss of an electron and the notation for the molecular ion are shown below.

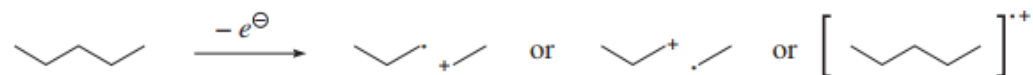
Loss of an electron from a non-bonding orbital:



Loss of an electron from a π orbital:



Loss of an electron from a σ orbital:



Principle of Mass Spectrometry – Summary

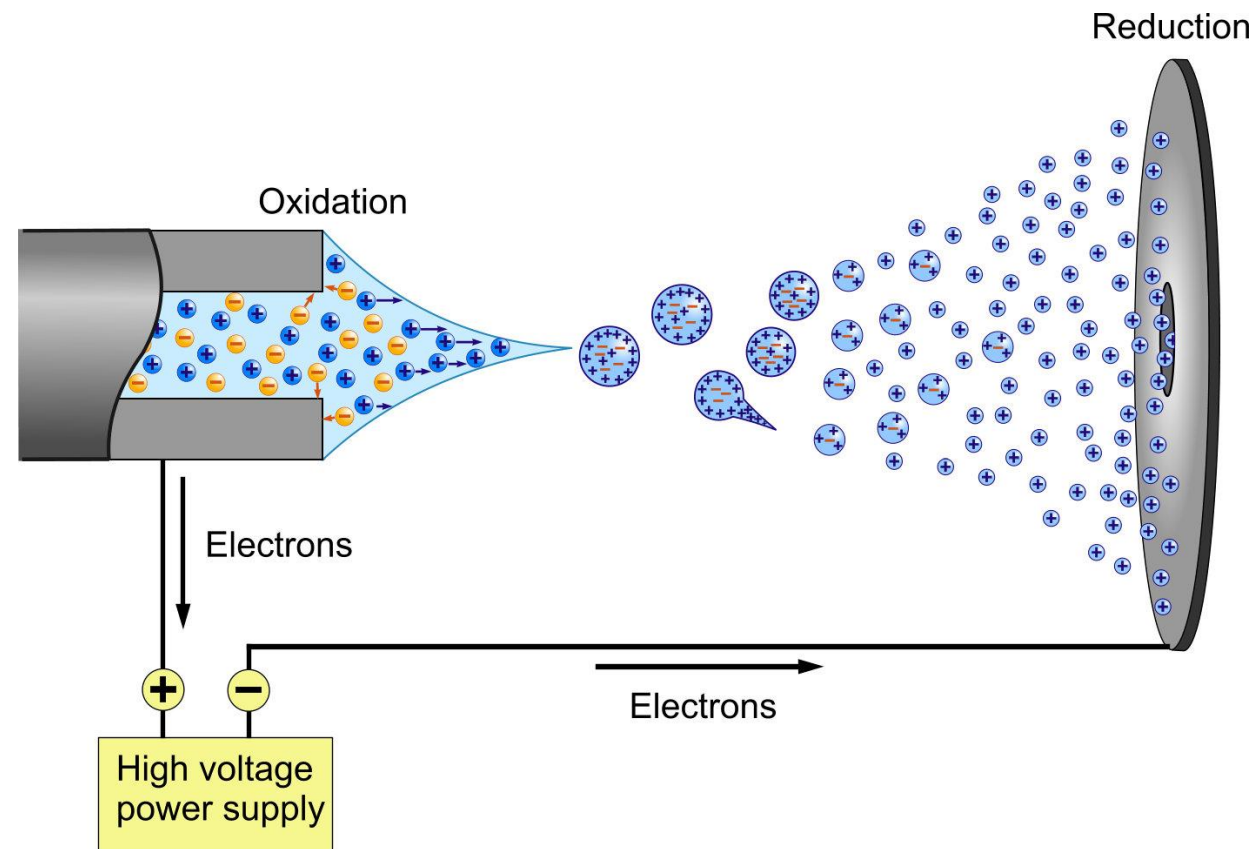
- Mass spectroscopy (MS) ionizes and energizes a molecule to determine its mass and connectivity through fragmentation.



- Electron impact- Mass Spectroscopy (EI-MS) uses high energy electrons to ionize the molecules.
- Only cationic fragments are responsive to the magnetic field and detected.
- Abundance of the ion detected is dependent upon their rate of formation and their rate of fragmentation.
- More stable ions that are easy to form tend to be responsible for more intense signals.

Electrospray Ionization (ESI)

- A softer technique than EI
- Especially useful for studying high molecular weight biomolecules and other labile or non-volatile compounds
- The solution of the sample is sprayed out the end of a fine capillary into a heated chamber (ambient pressure)
- A high voltage potential is applied across its surface; small, charged droplets are expelled into the ionization chamber
- The charged droplets are subjected to a counterflow of a drying gas that evaporates the solvent molecules from the droplets

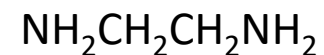
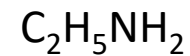


Determination of Molecular Weight and Molecular Formula

Nitrogen rule:

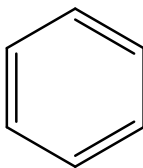
- Even or zero number of N atoms: molecular ion peak is even
- Odd N atoms: molecular ion peak is odd

Example:



☐ High resolution Mass Spectrometers

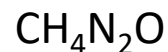
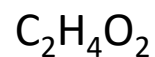
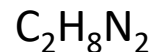
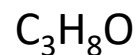
☐ Low resolution Mass Spectrometers



LRMS: 78

HRMS: 78.047

- HRMS allows a very precise mass determination and hence molecular formula



Molecular weight is 60.1



LRMS can't distinguish these molecules

The **Nitrogen Rule** is a fundamental principle in mass spectrometry used to determine the presence (or absence) of nitrogen atoms in an organic molecule based on its **nominal molecular mass**.

It is an incredibly handy "sanity check" when you are trying to narrow down a molecular formula from a mass spectrum.

The Core Principle

The rule is based on the unique property of Nitrogen (^{14}N): it is the most common isotope that has an **even atomic mass** (14) but an **odd valence** (3).

Here is how the rule breaks down for a molecule containing only C, H, N, O, S, or Halogens:

Nominal Molecular Mass (M^+)	Number of Nitrogen Atoms
Even	Zero or an Even number (2, 4, 6...)
Odd	An Odd number (1, 3, 5...)

Why Does It Work?

Most common elements in organic chemistry follow a specific pattern:

- **Carbon (^{12}C):** Even mass (12), Even valence (4)
- **Oxygen (^{16}O):** Even mass (16), Even valence (2)
- **Hydrogen (^1H):** Odd mass (1), Odd valence (1) — *Note: These "cancel out" in pairs to stay even.*

Nitrogen breaks the pattern. Because it has an even mass but connects to three other atoms (odd valence), every nitrogen you add flips the total mass of the molecule from even to odd, or vice versa.

However HRMS will

$\text{C}_3\text{H}_8\text{O}$ 60.05754

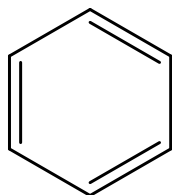
$\text{C}_2\text{H}_8\text{N}_2$ 60.06884

$\text{C}_2\text{H}_4\text{O}_2$ 60.02112

$\text{CH}_4\text{N}_2\text{O}$ 60.03242

➤ For the determination of molecular formula, HRMS is used

➤ However, there is a way by which we can use LRMS to figure out the molecular formula



Chemical Formula: C_6H_6

Exact Mass: 78.0470

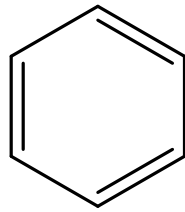
Molecular Weight: 78.1140

m/z: 78.0470 (100.0%), 79.0503 (6.5%)

(Chemdraw)

Isotope Ratio Data

- In mass spectrum, isotopic mass is measured; not the molecular weight
- We examine the peaks due to the molecular ion and related ions that bear one or more heavy isotopes (molecular ion cluster)



Chemical Formula: C₆H₆

Exact Mass: 78.0470

Molecular Weight: 78.1140

m/z: 78.0470 (100.0%), 79.0503 (6.5%)

PRECISE MASSES OF SOME COMMON ELEMENTS

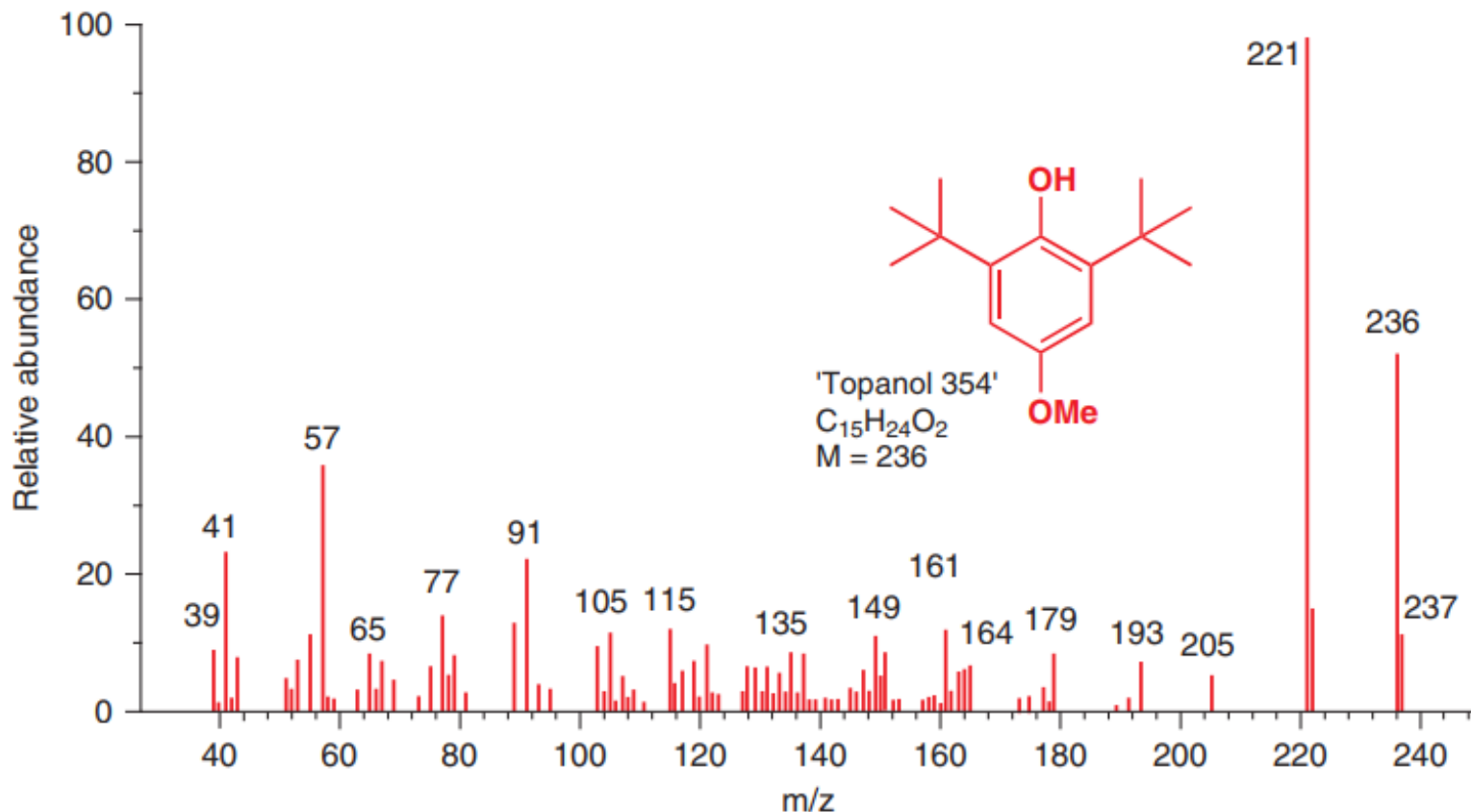
Element	Atomic Weight	Nuclide	Mass
Hydrogen	1.00797	¹ H	1.00783
		² H	2.01410
Carbon	12.01115	¹² C	12.0000
		¹³ C	13.00336
Nitrogen	14.0067	¹⁴ N	14.0031
		¹⁵ N	15.0001
Oxygen	15.9994	¹⁶ O	15.9949
		¹⁷ O	16.9991
		¹⁸ O	17.9992
Fluorine	18.9984	¹⁹ F	18.9984
Silicon	28.086	²⁸ Si	27.9769
		²⁹ Si	28.9765
		³⁰ Si	29.9738
Phosphorus	30.974	³¹ P	30.9738
Sulfur	32.064	³² S	31.9721
		³³ S	32.9715
		³⁴ S	33.9679
Chlorine	35.453	³⁵ Cl	34.9689
		³⁷ Cl	36.9659
Bromine	79.909	⁷⁹ Br	78.9183
		⁸¹ Br	80.9163
Iodine	126.904	¹²⁷ I	126.9045

Carbon has a minor but important isotope ^{13}C

The minor isotopes of many elements that appear at below the 1% level are not usually important, but one we cannot ignore is the 1.1% of ^{13}C present in ordinary carbon, of which the main isotope is of course ^{12}C .

The exact height of these peaks is useful as an indication of the number of carbon atoms in the molecule. Each carbon has a 1.1% chance of being ^{13}C rather than ^{12}C , so the more C atoms there are the bigger this chance becomes. If there are n carbon atoms in a molecular ion, then the ratio of M^+ to $[\text{M} + 1]^+$ is $100:(1.1 \times n)$.

Topanol 354: With 15 carbons, there's a 16.5% chance there will be one ^{13}C atom in the molecule, and you can clearly see the sizeable $\text{M} + 1$ peak at 237.



Q1. An unknown substance has a molecular ion peak at $m/z = 107$, with a relative intensity of 100. The relative intensity of the $M+1$ peak is 8.00. What is the molecular formula for this unknown?

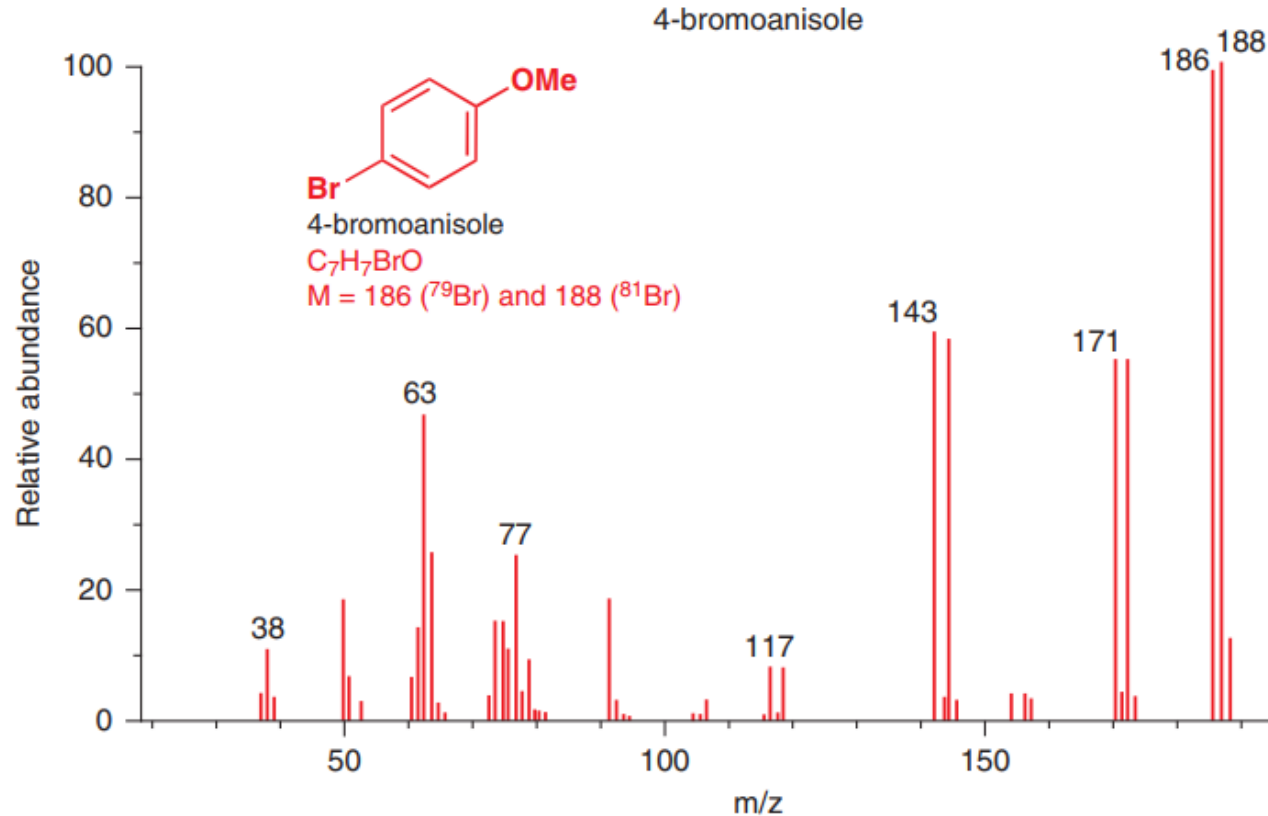
Mass Spectrometry Separates Isotopes

NATURAL ABUNDANCES OF COMMON ELEMENTS AND THEIR ISOTOPES

Element		Relative Abundance			
Hydrogen	^1H	100	^2H	0.016	
Carbon	^{12}C	100	^{13}C	1.08	
Nitrogen	^{14}N	100	^{15}N	0.38	
Oxygen	^{16}O	100	^{17}O	0.04	^{18}O 0.20
Fluorine	^{19}F	100			
Silicon	^{28}Si	100	^{29}Si	5.10	^{30}Si 3.35
Phosphorus	^{31}P	100			
Sulfur	^{32}S	100	^{33}S	0.78	^{34}S 4.40
Chlorine	^{35}Cl	100			^{37}Cl 32.5
Bromine	^{79}Br	100			^{81}Br 98.0
Iodine	^{127}I	100			

Mass spectrometry detects isotopes

Most elements can exist as more than one isotope. Usually, one isotope accounts for the vast majority (perhaps >99%) of the atoms of an element. But for some elements, atoms of several isotopes make up a significant proportion of the total in a sample. Chlorine, for example, is normally a 3:1 mixture of ^{35}Cl and ^{37}Cl , while bromine is an almost 1:1 mixture of ^{79}Br and ^{81}Br . Because mass spectrometry weighs individual molecules, there is no averaging: instead it detects the true weight of each molecule, whatever isotope it contains.



El mass spectrum

For a compound containing **two Bromine atoms**

Peak	Isotope Combination	Statistical Probability	Relative Intensity
M	$^{79}\text{Br} + ^{79}\text{Br}$	$0.5 \times 0.5 = 0.25$	1
M+2	$(^{79}\text{Br} + ^{81}\text{Br}) + (^{81}\text{Br} + ^{79}\text{Br})$	$(0.5 \times 0.5) + (0.5 \times 0.5) = 0.50$	2
M+4	$^{81}\text{Br} + ^{81}\text{Br}$	$0.5 \times 0.5 = 0.25$	1